

Cong Wang (王聪)

Postdoc, Peking University • frankcongwang@pku.edu.cn • 18811602976

Education

- Ph.D. in Physical Chemistry, Beijing Normal University, 2022
Advisors: Jiu-Shu Shao (邵久书), Jian Liu (刘剑)
- B.S. in Chemistry, Boling College (化学伯苓班), Nankai University, 2015
Advisors: Peng Cheng (程鹏), Wei Shi (师唯), Hong-Wei Sun (孙宏伟)

Research Experience

- Postdoc, School of Physics, Peking University, 2023 -
Advisor: Xin-Zheng Li (李新征)
- Research Assistant, Peking University, 2022
Advisor: Jian Liu (刘剑)

Publications

- [1] Wang†, He†, Zeng, Yang, Ye*, Li*, Revisiting the fermion sign problem from the structure of Lee-Yang zeros. III. Re-weighted simulation and identification of zeros.
- [2] He†, Zeng†, Yang, Wang*, Ye*, Li*, Revisiting the fermion sign problem from the structure of Lee-Yang zeros. II. Finite temperature results and continuation on the complex plane, arXiv: 2606.07415.
- [3] He†, Zeng†, Yang, Wang*, Ye*, Li*, Revisiting the fermion sign problem from the structure of Lee-Yang zeros. I. The form of the partition function for anyons and its zeros at 0 K, Phys. Rev. E, 113, 024115 (2026).
- [4] Liang†, Wang†, Wang†, Wang, She, Wang, Shao, Liu*, An efficient integrator scheme for sampling the (quantum) isobaric-isothermal ensemble in (path integral) molecular dynamics simulations, J. Chem. Theory Comput. 21, 6394 (2025).
- [5] Wang†, Zhang†, Liu*, Shao*, Generalized fourth-order decompositions of imaginary time path integral: Implications of the harmonic oscillator, Chin. J. Chem. Phys. 35, 516 (2022).
- [6] Li, Han, Chai, Wang, Zhang, Chen, Liu*, Shao*, Stationary state distribution and efficiency analysis of the Langevin equation via real or virtual dynamics, J. Chem. Phys. 147, 184104 (2017).
- [7] Wang, Zhang, Liu*, Shao*, An efficient algorithm for the fourth-order decomposition for path integral molecular dynamics. (In preparation)
- [8] Wang, Zhang, Liu*, Shao*, Application of the generalized fourth-order decomposition schemes to perturbed path integral. (In preparation)

Talks and Presentations

- Revisiting the fermion sign problem from the structure of Lee-Yang zeros, The 19th National Conference on Chemical Dynamics, September 18-21, 2025, Chun' an, Hangzhou, Zhejiang (Oral report)
- Fermion sign problem and distribution of Lee-Yang zeros, Frontiers in Ab Initio Quantum Molecular Dynamics (CECAM Flagship School), Jul 25-29, 2026, Beijing (Awarded poster)
- Lee-Yang zeros and path integral molecular dynamics of indistinguishable particles, The 35th Chinese Chemical Society Congress, Apr 11-14, 2026, Chongqing (Awarded poster)
- Efficient algorithms for sampling canonical and isobaric ensembles via molecular dynamics, The 31st Chinese Chemical Society Congress, May 5-8, 2018, Hangzhou, Zhejiang (Awarded poster)

Honors and Awards

- Functional Molecular Materials Research Group of Nankai University, Deng Xiaoping' s Scientific and Technological Innovation Team (小平科技创新团队), China Youth Science and Technology Innovation Award Fund, 2014 — Team leader

Research Interests

- Acceleration of path integral molecular dynamics (PIMD) methods. PIMD is an effective method for studying nuclear quantum effects, but its computational cost limits applications to biological and solvated systems. We developed the middle-scheme thermostat algorithm (adopted by software such as Amber) for sampling the canonical (JCP 2017) and isobaric-isothermal (JCTC 2025) ensembles; developed parameterized generalized fourth-order decompositions replacing second-order Trotter splitting, with parameters optimized on the one-dimensional harmonic oscillator (CJCP 2022); and proved that the perturbed path integral (PPI) is at least fourth-order accurate and can be combined with generalized fourth-order decompositions (to be submitted).
- Fermion sign problem and PIMD of indistinguishable particles. We connected the fermion sign problem (FSP) with the Lee-Yang theory of phase transitions and showed that the partition function is a polynomial in ξ with fixed zeros at zero temperature depending only on the particle number (PRE 2026). We explained the origin of the FSP with Lee-Yang zeros, analyzed the movement of zeros at finite temperature, and proposed analytic continuation from high-temperature thermodynamic properties (submitted to PRE). We proposed a numerical method to obtain Lee-Yang zeros of general interacting systems and to guide fermionic PIMD simulations (to be published). Together these three papers establish a theoretical framework for studying the FSP via Lee-Yang theory.

Future Research Plans

- Acceleration methods for path integrals of indistinguishable particles. Developing new coordinate transformations to decouple the coordinates of indistinguishable particles and improve sampling efficiency; combining our tunable generalized fourth-order decompositions with permutation blocking, higher-order fermionic propagators, and machine learning to expand the accessible system sizes and temperature ranges.
- Real-time dynamics methods for indistinguishable particles. Combining fermionic path integral methods with RPMD to compute dynamic structure factors of the uniform electron gas and tunneling splittings of methyl rotations in polymethyl systems, and with dissipative dynamics methods for reactions on surfaces and in cages, addressing problems in astrochemistry such as reactions on interstellar ice surfaces.

Referees

- Xin-Zheng Li (李新征) — School of Physics, Peking University — zxli@pku.edu.cn
- Jiu-Shu Shao (邵久书) — College of Chemistry, Beijing Normal University — jiushu@bnu.edu.cn
- Jian Liu (刘剑) — College of Chemistry and Molecular Engineering, Peking University — jianliupku@pku.edu.cn